



The Para-Athlete

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Participation in sports and physical activity by persons with different abilities continues to grow within the general athlete population and within the adaptive sports community.¹⁻⁴ Sports participation provides these athletes with benefits in terms of their general state of health, functionality, life skills, self-esteem, and overall quality of life.¹⁻⁴ Given the increase in participation, performance, and injuries among the adaptive athlete, the American College of Sports Medicine (ACSM) encourages managing exercise in patients with disabilities like the general population.⁵⁻⁷ It is incumbent upon medical providers to promote activity among this patient population but also to be knowledgeable and proficient regarding common complications of exercise to maximize their patients' safety and health.

This chapter provides clinicians with a broad overview of athletes with disabilities and important considerations regarding their optimal care. A brief overview of athlete competition classification systems and a discussion of the most common types of adaptive athletes are included, as well as any unique physiologic characteristics, medical needs, and adaptive equipment. Participation in the Paralympics is the pinnacle achievement of an adaptive athlete and is the primary context and framework for this discussion. However, the information and principles presented may be applied to nonelite athletes over a broad range of competition levels and sporting activities.

CLASSIFICATION

Athletes with disabilities are an inherently heterogeneous group with highly variable profiles of athletic capacity depending on the type, location, and severity of their inherent disability. Akin to the weight classes used in boxing and wrestling, the global governing body, the International Paralympic Committee (IPC), has developed classification systems. These classification systems maintain a measure of fairness in competition and most often apply to high-level athletes.⁸ The common purpose of the classifications systems in summary is to (1) define eligibility for participation and meet minimum disability criteria to participate, (2) ensure the athlete is competing against a class of athletes with impairments that cause a similar amount of activity limitation, and (3) ultimately allow for skill and fitness to determine success.^{8,9}

To be eligible for Paralympic sports, an athlete must have 1 of 10 eligible impairments (physical or structural): hypertonia, ataxia, athetosis, leg length difference, loss of muscle strength,

loss of passive range of movement, limb deficiency, short stature, visual impairment, or intellectual impairment.

Athletes must then meet the minimum eligibility criteria determined by each sport as defined by the regulations of the sport's governing body. For example, athletes with visual impairment do not meet the criteria to compete in Paralympic wheelchair tennis, whereas they do meet minimum criteria for multiple track and field sports, as well as many other sports.⁹

Lastly, athletes undergo a sports class allocation (SCA). SCA allows athletes with similar activity limitations to compete together, meaning the impairments of the athletes competing can often be different within a class. The sporting event greatly influences the classification process because different types and severities of disability affect performance to different levels in different sports. Therefore, depending on the sport and locale, designations may be condensed into fewer classes based on a wider range of disability. For example, within track and field, the disciplines and classes include: running and jumping (16 classes), wheelchair racing (7 classes), standing throws (15 classes), and seated throwing sports (11 classes). Meanwhile there are sports which may have one SCA, such as para powerlifting or wheelchair curling, in which once an athlete meets minimal eligibility criteria they are either sports class eligible or noneligible.⁹ Of note, team sports such as wheelchair basketball often incorporate a wide range of disability in direct competition via the use of point systems, in which athletes with less disadvantageous levels of disability are assigned higher point values, with the team not permitted to exceed a given total point value for its players in the game.

Direct observation of the athlete during play or competition is also a characteristic of some sports that are able to use more objective measures, such as perception and field of view in the visual impairment category.

Of the athletes who compete, there are six main major disability categories of athletes: amputees/limb deficient, wheelchair (typically spinal cord injury), cerebral palsy, visual impaired, the intellectually disabled, and "Les Autres." These athletes must meet 1 of the 10 eligibility requirements. Les Autres is a French term meaning "the others" and is made up of athletes with a variety of impairments and disabilities. Major disability category rather than IPC impairment-based classification will be used going forward to discuss differences in athletes with regard to exercise physiology, medical considerations, and adaptive sports equipment.

Abstract

Sports and physical activity participation by persons with different abilities is an important aspect of sports medicine. This chapter provides clinicians with a broad overview of athletes with disabilities and important considerations regarding their optimal care. This includes discussion on athlete competition classification systems, the most common types of adaptive athletes, and the unique physiologic characteristics, medical needs, and adaptive equipment for these athletes.

Keywords

wheelchair
intellectually disabled
visually impaired
spinal cord injury
amputee
cerebral palsy
adaptive equipment

ATHLETES WITH LIMB DEFICIENCY/ AMPUTEE ATHLETES

Limb deficiency may be congenital, illness related, or traumatic and may involve the upper and/or lower extremities. The athlete with limb deficiency may compete as a standing or seated competitor depending on the sport and the participant's level of limb loss. Given the variance of the location and length of the congenitally abnormal or surgically amputated extremity, these athletes have a wide range of functionality.

Upper extremity limb deficiency is most commonly from amputation caused by trauma, followed by cancer, and then vascular disease.¹⁰ In 2005 it was estimated that almost 41,000 were living with upper extremity limb loss.¹¹ The level of amputation is key in upper extremities for function and range of motion (ROM), as the preservation of movement in both the elbow and forearm is essential for positioning the hand in space and therefore performance of function.¹² The most common level of upper extremity amputation is transradial.¹⁰

Lower extremity limb deficiency is often secondary to amputation from peripheral vascular disease (82%), followed by trauma (16%), malignancy (0.9%), and lastly congenital deformity (0.8%).¹³ The vast majority of lower limb amputations are at the transtibial and transfemoral levels.^{10,13}

Given that the most common limb-deficient athlete is an amputee, amputee literature will be predominately referenced going forward.

Exercise Physiology

It is presumed that the cardiopulmonary physiology of amputee athletes with traumatic amputations or with vascular disease (dysvascular amputations) is similar to that of able-bodied athletes. Some exceptions may exist for persons who have had an amputation for congenital reasons when a cardiac abnormality may be a component of a syndrome.

The amputation of a lower extremity does however significantly affect the “energy expenditure” used during ambulation. Various measurements have been used to quantify the “expenditure” of ambulation in persons with limb loss, and these calculations can be expressed in functions of distance, rate, and velocity. Persons with limb loss typically walk slower than non-amputees to maintain a similar oxygen consumption rate. Therefore in athletic competition the amputee athlete will need to increase his or her rate of oxygen consumption to maintain a similar velocity of a nonamputee athlete. This energy use also differs between vascular versus nonvascular amputees of the same level of amputation (Table 28.1).¹⁰

Athletes with amputations or congenital limb abnormalities can compete in a variety of sports with various permissions for adaptive equipment, which may allow the use of a prosthesis, or the athlete can compete in certain wheelchair sports.¹⁴

Medical Considerations

Musculoskeletal Injuries

Injury patterns among amputee athletes are similar to those for athletes without disabilities; however, the severity and location of the missing limb dictates the frequency and type of injury.

TABLE 28.1 Energy Expenditure of Ambulation

Level of Amputation	Etiology	Unilateral/ Bilateral	% of Energy Increase per Unit of Distance
Transtibial	Traumatic	Unilateral	25% (short: 40%; long: 10%)
		Bilateral	41%
Transfemoral	Dysvascular	Unilateral	40%
	Traumatic	Unilateral	60%–70%
Transfemoral + transtibial	Dysvascular	Unilateral	100%
		Bilateral	118%

From Uustal H, Baerga E, Joki J. Prosthetics and orthotics. In: Cuccurullo SJ, ed. *Physical Medicine and Rehabilitation Board Review*. Demos Medical; 2015.

Lower extremity amputees are at risk for injuries in both intact and residual limbs. The distal residual limb is often the site of skin trauma due to the prosthesis. Altered biomechanics, because of the asymmetric need for power and propulsion, may predispose athletes to knee, hip, sacroiliac, and lumbar spine pain.^{15–17} Alternatively, an amputee may rely too heavily on the intact limb, which can increase the risk for overuse injuries and osteoarthritis.^{18–21}

Similarly, in upper extremity amputees, altered use and load patterns of the intact limb can result in pain and injury to either limb or axial structures. Overuse injuries such as shoulder impingement, rotator cuff tears, epicondylitis, and peripheral nerve entrapments are common in the intact limb of upper extremity amputees.^{22–24} Differences in weight and swing excursion between the upper extremities, as well as the need to compensate at the shoulder for loss of distal joint function, may cause significant asymmetry in the demands of the thoracic and cervical spines and paraspinal and parascapular musculature, resulting in pain and dysfunction.^{25,26} Treatment for many of these conditions is identical to that for nonamputees. In addition, however, precise fitting of prostheses in this population is a critical aspect of beneficial treatment.

Dermatologic Conditions

After amputation of the limb and subsequent fitting of a prosthesis, the skin of the distal portion of the residual limb becomes a weight-bearing surface and as a result is at increased risk for skin disorders. Although studies can vary, studies range from 40% to 73.9% of amputees with at least one skin problem.^{27–29} Risk factors that have been identified are male sex, older age, and amputation from diabetes or peripheral vascular disease.²⁸

Rashes are frequently observed in persons who use prostheses. A noninfectious allergic rash should trigger examination of the type of liner used, with the goal of incorporating material that is less irritating and that promotes greater perfusion of sweat away from the skin. An allergen (including detergents, lotions, or the liner material) has been the cause in up to 43% of patients with dermatitis.²⁷ Cleaning the residual limb and

prosthesis regularly can prevent rashes. Prevention of infections is a key priority and often a source of concern. Folliculitis, boils, and abscesses should be treated appropriately, along with keeping the skin dry and clean. As such, a break from the prosthesis may be warranted.¹⁰ Keep in mind that the rash could also be fungal, which is commonly seen in excessive sweating. The rash would need to be treated, but long-term solutions to decrease the sweating by using an antiperspirant can be considered.¹⁰

Verrucous hyperplasia is a wartlike lesion that may develop at the distal end of the residual limb. It may occur as a result of untreated proximal residual limb constriction (choke syndrome) from a socket or wrap that has caused decreased pressure in the distal residual limb. Edema formation occurs and over time will result in wartlike skin overgrowth. The reversal and treatment of verrucous hyperplasia consists of equal distribution of pressure through the residual limb by relieving proximal constriction and reestablishing total contact within the socket.¹⁰

Blisters and sores are very common and can occur as a result of friction between skin of the residual limb and the prosthesis. Skin breakdown may also occur when pressure is applied disproportionately to a pressure-sensitive area of skin on the residual limb, such as the tibial tubercle in a transtibial amputee.¹⁰ The risk may be compounded by impaired sensation in the residual limb or sweating with athletic activity, which may increase moisture at the skin-socket interface and heighten the risk for skin breakdown. A properly fitting socket will normally prevent blisters and sores, and it is important to maintain a good socket fit by adding and removing socks throughout the day to adjust for volume reduction,

Heterotopic Ossification

The formation of bone in tissues that are not normally ossified, heterotopic ossification (HO), usually evolves after traumatic brain injury, spinal cord injury, burns, and total arthroplasty. Recently, HO has been reported to develop frequently in injured tissue of residual limbs in traumatic amputees,³⁰ which may increase risk of skin breakdown or stimulate pain with weight bearing. HO is also more likely to develop around joints and muscle adjacent to trauma and typically develops within the first 6 months to a year after amputation. The development of HO typically occurs during the time an amputee is beginning prosthetic training. Thus the majority of persons with HO are diagnosed prior to engaging in athletic competition, which facilitates modifications in design of the socket and vigilance for signs of skin breakdown. Treatments for HO can include ROM, physical therapy, radiation, surgery, and medications.³¹ Treatment options vary with location of the HO and medical comorbidities of the patient; therefore treatment should be tailored to each patient specifically.

Bone Spurs

Bone spurs and overgrowth can occur in cases where the periosteum of the bone is stripped during surgery or from trauma. The bone grows faster than the overlying skin. Bone spurs are more common in kids with acquired amputation and in young adults with traumatic amputations.¹⁰

Neuroma

Neuromas form at the distal end of transected nerves in the residual limbs of amputees. The development of a neuroma in a weight-bearing structure or in its vicinity can cause severe pain with ambulation and weight bearing and limit an athlete's ability to train and compete. Treatments include prosthetic modification to relieve pressure; oral medications, including antiepileptic agents and tricyclic antidepressant drugs (neuropathic medications); injections of corticosteroids and local anesthetics into the neuroma; and radiofrequency ablation of neuromas.^{32–34} Many common medications that are used to treat neuropathic pain may be restricted in competition by the World Anti-Doping Agency (WADA), and clear knowledge regarding these limitations is vital for medical practitioners. We encourage medical practitioners who are caring for athletes with disabilities to visit WADA's website to review approved and restricted medications "in and out of competition."³⁵ Surgical excision, which can improve pain and patient quality of life, may ultimately be necessary if conservative treatments fail.^{36–38}

Adaptive Sports Equipment

Numerous sporting events incorporate adaptive sports equipment for amputees. It is beyond the scope of this chapter to discuss the numerous prosthetic modifications and other adaptive equipment for several sports and recreational activities. However, medical practitioners should consider several pertinent facts when they prescribe adaptive equipment. Compared with common prosthetic devices, a range of modifications should be considered in this population.³⁹ Of particular importance is the prosthetic weight, particularly in sports in which increased weight may negatively affect speed. Occasionally, use of a conventional prosthesis may be advantageous compared with a technologically advanced prosthesis. The clinician should also consider alignment, prosthetic foot dynamics, shock absorption, and the possible need for transverse rotation.

Remember, not all athletes with limb deficiency will need or use a prosthesis to compete in their sport. For example, in para table tennis, athletes may use tape to hold their paddle in place on their limb. Athletes with limb deficiency can also use wheelchairs or Lofstrand crutches, depending on the sport.

WHEELCHAIR ATHLETES

Wheelchair athletes may have spinal cord injuries, amputations, or neurologic disorders such as polio, spina bifida, and cerebral palsy. The majority of wheelchair athletes have spinal cord injuries; hence discussion in this section is limited to the nuances of athletic participation of athletes with spinal cord injuries.

Exercise Physiology

The exercise capacity and physiologic responses to exercise of wheelchair athletes are different from that of able-bodied athletes. Furthermore, inherent physiologic differences among wheelchair athletes are observed. Based on the level and severity of their injury, their maximal exercise capacity may range from being comparable with either able-bodied athletes or sedentary able-bodied persons.

Spinal cord injuries and other neurologic pathologies induce a varying degree of paralysis of voluntary muscles, with a resultant decrease in muscle mass available for exercise. Diminished muscle mass also has a negative impact on performance via impaired supporting dynamic restraints (such as core musculature) and impaired hand and arm function in tetraplegic athletes. These phenomena can reduce the efficiency of energy transfer from the athlete to the chair and other objects. However, these differences in muscle mass only partly explain differences in exercise capacity, because the arteriovenous oxygen difference in submaximal exercise is typically altered in wheelchair athletes compared with able-bodied athletes.⁴⁰ This difference may be indicative of different levels of muscle recruitment to achieve a given work rate, as well as impairments in local and regional blood flow in response to exercise. In general, the lower the spinal level of the lesion, the more muscle mass is available for exercise, which translates to improved vasomotor regulation. Thus power output is inversely related to the level of the spinal lesion. Of note, in this population, high values of aerobic and anaerobic output which normally correlate with performance, may not correlate in this population.⁴¹ For example, wheelchair tennis players with low-level spinal cord injuries have been able to reach physiologic thresholds for exercise intensity similar to able-bodied tennis players,⁴² but these parameters (peak oxygen uptake) did not correlate to rank and performance.⁴³

Cardiovascular responses to exercise are also altered in comparison with able-bodied athletes and demonstrate significant differences between wheelchair athletes with higher and lower level lesions. The loss of vasomotor regulation and active muscle pumping action below the level of the lesion results in impaired venous return, thus restricting central blood volume. Impaired sympathetic innervation to the heart, which is absent in persons with complete spinal cord lesions above the T1 level and reduced in persons with complete spinal cord lesions above the T6 level, results in a blunted cardiac response, and maximal heart rate is limited to 110 and 130 beats per minute via intrinsic sinoatrial activity.^{41,44,45} Spinal cord athletes with cervical level injuries also demonstrate attenuated cardiac size and left ventricular hypertrophy compared with those with lower level injuries.⁴⁶ These factors lead to impaired cardiac performance, with reductions in cardiac output and stroke volume up to approximately 30% in persons with motor-complete cervical spinal cord injuries.^{47,48} Although complete motor cervical spinal cord injuries should have no intact sympathetic nervous system, there may be a degree remaining, and this can vary between athletes. A 2013 study demonstrated in wheelchair rugby athletes with motor complete cervical injury who had partial preservation of descending sympathetic response (demonstrated by skin sympathetic response [SSR]), SSR correlated to heart rate peak and the degree of preservation further correlated to performance.⁴⁹

With regard to heart rate response to exercise to measure fitness, it is likely more reliable to compare ratings of perceived exertion to monitor training and fitness improvements in SCI athletes.⁵⁰ For example, cerebral palsy athletes participating in power soccer had a significantly higher (>12 beats/min) heart rate response compared with spinal cord athletes. Furthermore, 22 athletes with cerebral palsy and 5 with muscular dystrophy

had heart rates that exceeded 55% of the estimated maximum heart rate for at least 30 minutes of competing, compared with only 1 in the spinal cord injury group.⁵¹

Pulmonary function is also often impaired and influenced by the level of the lesion. These deficits occur primarily via paralysis of the expiratory muscles; however, persons with cervical lesions may also have paralysis of inspiratory musculature, resulting in significant impairments in ventilation relative to paraplegic persons.⁵² Compared with able-bodied persons, the forced expiratory volume in 1 second, forced vital capacity, and tidal volume are approximately 90% in paraplegic patients and 60% in tetraplegic patients.⁵³

Medical Considerations

Musculoskeletal Injuries

Wheelchair athletes are at risk for routine musculoskeletal injuries, such as muscle strains, at rates comparable with those of able-bodied athletes. Injuries in this population may affect participation and performance to a disproportionately high degree compared with able-bodied athletes, depending on the sport.

Upper extremity injuries are more common than lower extremity injuries in wheelchair athletes. The adaptive sport literature suggests that injuries most commonly involve the shoulder, ranging between 15% and 72%.^{6,54} Arm and wrist injuries follow with the most likely diagnosis being muscle strains, tendinopathy, bursitis, and contusions.^{55,56}

A history of shoulder pain is reported in more than 90% of selected wheelchair athlete populations, with an increasing prevalence in proportion to the amount of trunk and upper extremity disability.^{54,57,58} Shoulder impingement syndrome is the most common injury; bicipital and rotator cuff tendinopathy and tears are also common.⁵⁹ Wheelchair athletes often have a protracted scapular position with dynamic scapular dyskinesis, resulting in loss of subacromial space and impingement.⁶⁰ In addition, the humeral head is often elevated as a result of relative muscle strength imbalance, favoring shoulder abduction strength over adduction and rotational strength. This risk can potentially be modifiable by alterations in trunk inclination, backrest height, and the position of the wheelchair axle relative to the shoulder. Physical therapy should improve shoulder flexibility, correct scapular kinematics, and address shoulder muscle imbalances.^{61,62} Much attention has reviewed the “lack of the kinetic chain” and its impact on shoulder injuries. It appears this may be sport dependent, and, most importantly, the lack of kinetic chain should not be assumed to be the reason for pathology.^{63,64}

Lower extremity injuries are also seen in this population, although they are less frequent than in ambulatory athletes.⁶⁵ Lower extremity fractures are more common in high-speed adaptive sports with collision, such as wheelchair basketball, rugby, and softball.⁶⁶ Compared with upper extremity injuries caused by overuse, these injuries are usually acute and posttraumatic. There are high rates of osteoporosis observed in lower extremities in the spinal cord-injured population. Thus clinicians should be vigilant because even minor trauma may result in fractures, and any resulting deformity or angulation after healing may adversely affect available seating positions, increase the risk for pressure sores, and alter functional status.

Peripheral Nerve Entrapment

Peripheral nerve entrapments are commonly encountered in wheelchair athletes. Significant entrapment neuropathies include median neuropathy at the wrist, ulnar neuropathy at the elbow, and radial neuropathy, likely partly secondary to the biomechanics required for wheelchair use.^{67–69} Shoulder muscle imbalance and scapular dyskinesis may also result in suprascapular neuropathy at either the suprascapular or spinoglenoid notches.⁷⁰ In addition to standard therapies, protection of the wrist and elbow with padded gloves and sleeves, use of a wheelchair that has been properly fit to the athlete, and use of an adequate ROM and proper techniques during propulsion may help with prevention and treatment.^{67,71}

Thermoregulation

Altered sympathetic, vasomotor, and sudomotor responses and diminished rates of venous return from the loss of muscle-pumping action result in a diminished ability to sense and respond to thermal imbalance.^{72,73} The loss of skeletal muscle and function also impairs the response to negative thermal imbalances because of a reduced capacity for shivering.⁷⁰ These factors increase the risk for hypothermia and exertional heat illness, especially in persons with spinal cord lesions above the T6 level.⁷⁴

Pressure Sores/Ulcers

Pressure sores are common among wheelchair athletes, particularly those with a spinal cord injury. Prolonged pressure, typically over bony prominences such as the sacrum and ischial tuberosities, combined with insensate skin that is moist from activity, increase compressive and shears forces, resulting in local tissue ischemia and injury. Wheelchair athletes with altered trunk stability often maintain a flexed lower extremity posture at the hips and knees that further distributes the athlete's weight to "at-risk sites."⁷⁰ Vigilance is mandatory for pressure sore prevention, including skin checks, shifting of weight every 15 to 20 minutes to relieve pressure, the use of appropriately fit seat cushions, and maintenance of a dry environment.

Spasticity

Spasticity is common in wheelchair athletes who have spinal cord injuries and is manifested by involuntary muscle contraction, hyperreflexia, and velocity-dependent increases in tone. Synergistic muscle activation patterning may inhibit performance by altering positioning within the wheelchair, item control/accuracy, and propulsion. However, it is important to note that in some cases, spasticity may be beneficial for these functions when other voluntary muscle action is insufficient. It is important for the sports physician to remember that spasticity is primarily a velocity-dependent increase in tone, and thus a routine bedside examination may not be conducive to a thorough appreciation of the detrimental effects of spasticity unless sports activities are entirely replicated.

Spasticity is usually treated with oral medications, physical therapy that emphasizes ROM, and tone-reducing orthoses. In more advanced cases or in cases in which the athlete cannot tolerate the sedative effects of antispasticity medications, Botox

injections can focally reduce high levels of spasticity in a dose-dependent fashion. Surgical approaches such as muscle/tendon lengthening, tendon transfers, and selective dorsal rhizotomy may be considered in refractory cases.

Neurogenic Bladder and Bowel

Bladder dysfunction is common in wheelchair athletes who have spinal cord injuries and spina bifida. This dysfunction can result in urinary retention, often necessitating indwelling, suprapubic, or intermittent catheterization and resulting in increased rates of urinary tract infections and stone formation. Clinical presentation may include fever, fatigue, a general sense of unease, discomfort in the area of the bladder and kidneys, autonomic dysreflexia (AD), incontinence, and an increased level of muscle spasticity.

Bowel dysfunction is also common in persons with spinal cord injuries and spina bifida. Injuries above the S2–S4 spinal cord segments results in a "reflexic" neurogenic bowel, whereas injuries distal to these segments or involving the destruction of anterior horn cells (at these levels) result in an "areflexic" bowel. In addition to using other methods of facilitation, these persons are counseled to initiate regular bowel programs at the same time of day to condition the bowel to achieve effective defecation and avoid incontinence.

In the sports setting, the athlete may not adhere to appropriate frequency or technique of clean intermittent catheterization or appropriate timing of their bowel program because of an over focus on training, which may propagate infection, reduce vigilance of hydration status, and increase the risk of incontinence. Clinicians should counsel athletes on the importance of bowel and bladder management, which has significant performance and practical implications. Athletes should also be counseled to undergo slow alterations in bowel management to best accommodate their expected competition schedule at upcoming events and to always be cognizant regarding the available facilities for bowel and bladder management at each venue.

Heterotopic Ossification

HO may occur in up to 53% of patients with spinal cord injuries.⁷⁵ The development of HO is most common in the first 2 to 3 weeks after spinal cord injury. HO most commonly affects the hips but may also affect the knees, shoulders, and elbows.⁷⁵ HO may limit function and performance by restricting ROM for different seating positions, propulsion, and other tasks; it also may increase the risk for pressure sores and nerve entrapments.

Autonomic Dysreflexia

AD is a medical emergency usually seen in persons with complete spinal cord injuries involving or above the T6 segment. Noxious stimuli below the level of injury can cause reflexive sympathetic activity that cannot be modulated by supraspinal centers of control, resulting in high levels of sympathetic activity below the level of injury and incomplete parasympathetic compensation above the level of injury.⁷⁶

During an episode of AD, persons typically experience hypertension and headache due to vasoconstriction below the level

of injury, along with skin flushing, piloerection, diaphoresis, and bradycardia above the level of injury. Although mortality is rare, significant morbidity may result, including cerebral hemorrhage, seizures, and myocardial infarction. Treatment should include sitting the person upright, removing restrictive clothing, and searching for the source of the noxious stimulus, which is commonly a distended bladder or impacted colon, pressure sores, or another injury.⁷⁷ Systolic pressures of 150 mm Hg should be treated with short-onset and half-life antihypertensive medications. Transdermal nitro paste is highly effective and can be removed easily after resolution of the episode of AD to avoid subsequent hypotension.

An unsettling trend among wheelchair athletes is the intentional use of AD to improve performance during competition, a phenomenon known as “boosting.”^{78,79} Athletes will create a self-induced noxious stimulus via methods such as kinking a bladder catheter, sitting on ball-bearings, fracturing a toe, or excessively tightening legs straps to induce AD.⁷⁹ This strategy may lead to improvements in peak heart rate, peak oxygen consumption, and blood pressure and has been shown to enhance performance by 7% to 10%.^{78,80,81} For obvious health and safety reasons and to promote fairness during competition, the IPC has banned the use of induced AD from competition and has begun testing athletes prior to competition at Paralympic events.⁸²

Adaptive Equipment

Wheelchairs used in sporting endeavors are typically designed specifically for the particular demands of the sport involved, such as speed and endurance for racing, rigidity and stability for throwing, or agility and ruggedness for wheelchair basketball and rugby. Several variables can be manipulated to meet these needs: axle position (horizontally and vertically), wheels (number, size, and camber), hand rim size, seat back height and angle, seat cushioning, footplate (height and angle), the location and types of belts, and other fixations. For example, an elevated seat height may be advantageous for sports events such as wheelchair basketball, whereas a lower seat may be advantageous for sports requiring extra stability. The particular needs of the athlete and his or her disability level must be considered, including the level of truck support, altered location of the center of mass, and impaired hand function. These considerations are not only important to maximize performance but also to reduce the risk of injury and medical complications. Consultation with physical and occupational therapists and with prosthetists may be necessary depending on the needs and goals of the athlete.

ATHLETES WITH CEREBRAL PALSY

Cerebral palsy is defined as a group of disorders of the development of movement and posture that cause activity limitation and are attributed to nonprogressive disturbances that occurred in the developing fetal or infant brain.⁸³ Cerebral palsy is often accompanied by disturbances of sensation, cognition, communication, perception, and/or behavior, along with seizure disorder.⁸³ Patients with cerebral palsy are clinically classified on the basis of the type of movement disorder and the anatomic distribution. Cerebral palsy often shares clinical patterns similar

to those that occur with stroke and traumatic brain injuries. Consequently, athletes with impairments resulting from stroke or traumatic brain injury are often competing against cerebral palsy athletes.

Exercise Physiology

Studies assessing cardiorespiratory endurance, as measured by peak oxygen consumption during arm or leg ergometry, have consistently shown lower values for persons with cerebral palsy when compared with able-bodied control subjects.^{84–90} Similarly, compared with able-bodied control subjects, a cerebral palsy group showed significantly lower muscular endurance measured with a hand grip dynamometer.⁹¹ Muscular strength also appears to be decreased in persons with cerebral palsy when measuring force production in the lower extremities.^{91,92} Of note, a 2015 study did demonstrate similar fatigue in elite cerebral palsy athletes compared with able body athletes, despite a significant difference in power.⁹³ This may suggest that previous studies not performed in the elite cerebral palsy population must be taken with that limitation in mind.

Factors that may cause decreased muscular strength and endurance include desynchronization of agonist/antagonist muscle groups, hypersensitive tonic stretch reflexes, an increase in the ratio of type I versus type II muscle fibers, and reduced muscle volume.^{90,94} These factors may also account for the increased cost of energy and higher physical strain observed during ambulation in persons with cerebral palsy compared to able-bodied controls.^{88,95} An equivalent cost of energy has been shown in persons with acquired traumatic brain injury and survivors of a stroke.^{96,97}

The ACSM guidelines for persons with disabilities, including cerebral palsy and persons with similar chronic disorders, are detailed in [Box 28.1](#).⁹⁸ Some authors believe that persons with cerebral palsy need to maintain higher levels of physical fitness than the general population to offset the decline in function that occurs as a result of aging, including changes related to their condition.⁹⁰ It has been shown that persons with cerebral

BOX 28.1 American College of Sports Medicine Exercise Recommendations for Persons With Chronic Diseases and Disabilities Such as Cerebral Palsy

1. Professionals trained in strength and conditioning should supervise training exercise.
2. If there are significant movement limitations, obtaining moderate to vigorous exercise may be difficult, and reducing sedentary behaviors may be more reasonable.
3. For persons with mild to moderate impairment, aerobic capacity training is recommended in form of walking over ground or treadmill, or using ergometer or bicycle.
4. Perform single joint movements during training, one side at a time, and train these muscles within their available active range of motion.
5. Train the movement, prior to weight training and train power within the muscles available range.
6. Using a 1RM max calculator via the 10-repetition maximum is most feasible.

palsy respond well to resistance training and are capable of attaining normal strength.⁹⁹ It is also important to note that most studies in cerebral palsy persons in wheelchairs (that focus on physical activity and fitness interventions) indicate exercise is safe and promotes health, fitness, and well-being.¹⁰⁰ However, there are many types of exercise these patients can participate in and various sports may impact the success of exercise. The Cerebral Palsy International Sports and Recreation Association (CP-IRSA) is therefore helping to lead the world of research for exercise in cerebral palsy and related conditions, by studying the impact of training techniques, equipment, and impairment classifications in individual sports.

Medical Considerations

Medications

In persons who have had a traumatic brain injury, neurostimulants such as modafinil may increase susceptibility to either arrhythmia or heat exhaustion. Anticholinergic medications, which are used frequently by this population for the treatment of neurogenic bladder, can also increase susceptibility to heat exhaustion.¹⁰¹ Medications used commonly in this population may be banned by WADA and require therapeutic use exemption forms. Individuals and clinicians can obtain therapeutic use exemption forms at the WADA website.³⁵

Musculoskeletal Injuries

Musculoskeletal injuries are frequent in persons with cerebral palsy. Muscle strains and soft tissue injuries are most common.²⁴ The most frequent location includes the knee, followed by injuries to the shoulder.¹⁰² Muscular imbalances and spasticity produce increased tension across the knee that can trigger patellofemoral pain syndrome, chondromalacia, and chronic knee pain. Hip development is typically poor in persons with cerebral palsy and can cause acetabular dysplasia and hip subluxation.¹⁰³ Authors have suggested that tone, which may limit power production, may lead to increased joint stability.^{14,93}

Wheelchairs are used by approximately half of all athletes with cerebral palsy, which can lead to a high prevalence of shoulder injuries.^{24,103} Injuries resulting from wheelchair use have been reviewed in the “Wheelchair Athlete” section of this chapter. Musculoskeletal rehabilitation in athletes with cerebral palsy should follow the same general principles used in able-bodied persons but with a special emphasis on the recognition of muscular imbalances that make this population more susceptible to injury.⁴¹

Epilepsy/Seizures

Epilepsy affects approximately half of all patients with cerebral palsy and has numerous ramifications.¹⁰⁴ Exercise in epileptic patients improves aerobic performance and sense of well-being, reduces sleep disorders and fatigue, and decreases seizure occurrence.¹⁰⁵ It is very important to support exercise, rather than restrict it, whenever deemed safe. The International League against Epilepsy reports that individuals who are seizure-free for 1 year should be cleared for most activities.¹⁰⁵ With regard to collision and contact sports, with appropriate protection, patients with epilepsy should not be restricted. With regard to water sports (except

deep diving), athletes with good control can participate with direct supervision. Hang gliding, parachuting, and free climbing should be restricted due to the risk a seizure and subsequent fall poses. Gymnastics may be considered with direct supervision.¹⁰⁵

Antiepileptic drugs may have a sedating effect in persons with cerebral palsy, traumatic brain injury, or stroke because they already have an inherent risk for diminished arousal and concentration. Furthermore, antiepileptic drugs may contribute to the risk for thermoregulatory impairment, impaired vision, and decreased bone mineral density. In addition to antiepileptic drugs, a lack of mobility, poor feeding, and nutritional deficiencies can contribute to osteopenia. Seventy-seven percent of persons with moderate to severe cerebral palsy are found to have reduced bone mineral density, as measured in the femur.¹⁰⁶ As a result of a higher risk of fractures with osteopenia, athletes with cerebral palsy should have their bone mineral density assessed, especially before commencing participation in a contact sport.¹⁰⁷

Spasticity

Spasticity is common in the cerebral palsy population. The “Wheelchair Athlete” section of this chapter includes information regarding the recognition and treatment of spasticity. It is important to recognize that using certain medications can change an athlete’s classification but are not banned from competition when needed. Spasticity can either improve or impair performance.¹⁴

Adaptive Equipment

As previously mentioned, wheelchair use is conspicuous among athletes with cerebral palsy. The “Wheelchair Athlete” section of this chapter includes a detailed discussion of the use of wheelchairs by paraplegic athletes. Most stroke survivors and many patients with cerebral palsy or a traumatic brain injury are hemiplegic rather than paraplegic. Depending on the sport, hemiplegic athletes may benefit from an ultra-lightweight chair that is lower to the ground than the typical wheelchair, with the footrest of the unaffected side eliminated to allow for steering and propulsion with the unaffected lower limb. In persons with severe motor deficits who need greater trunk control, adaptive power chairs, seating, and pelvic restraints are essential to facilitate athletic participation. There are also restrictions such as that wheelchairs during racing may not have any mechanical gears or levers.¹⁴ Ambulatory athletes with cerebral palsy may benefit from wearing ankle-foot orthotics, which have been shown to normalize ankle kinematics in stance, increase stride length, decrease cadence, and decrease the energy cost of walking.¹⁰⁸ When prescribing ankle-foot orthotics, the suitable fit of an orthosis in specialized athletic shoes (e.g., track spikes or soccer cleats) should be considered. Because ataxia, seizures, and delayed reaction times are common in patients with cerebral palsy, helmets and mouth guards should be supplied to athletes at risk for trauma from falls or an impact from other players or sports implements.¹⁰⁵

INTELLECTUALLY DISABLED ATHLETES

The intellectually disabled athlete may include athletes with a wide array of underlying conditions. The current literature

predominantly pertains to persons with Down syndrome, but conditions such as fragile X syndrome and autism spectrum disorders are also common. To qualify as intellectually disabled for the Paralympic Games, a person must meet the following criteria¹⁰⁹:

1. Have an IQ score of 75 or lower, adaptive behavior, and age of onset younger than 18 years of age
2. Verification process of the evidence from #1 above
3. Reduced number of tests to measure IQ as a part of #1 above

Once an athlete meets the disability criteria, they will undergo on-site verification of sport-specific eligibility with a battery of tests and possibly sports-specific performance as well.^{9,109}

Criteria for participation with other organizations for the intellectually disabled have similar but more inclusive criteria to allow for increased participation by nonelite athletes. For example, the Special Olympics requires the participant to be at least 8 years old and possess an intellectual disability, cognitive delay, or developmental disability that results in functional limitations in general learning and adaptive skills, regardless of severity, as documented by an agency or professional.¹¹⁰

Exercise Physiology

Although responses to training are typically similar, differences in magnitude of response and overall capacity may exist. Persons with Down syndrome typically have lower peak oxygen uptake, maximum heart rate, and maximum ventilation compared with nondisabled control subjects and intellectually disabled persons without Down syndrome.^{111–113} Acute cardiac response and recovery from exercise also appear to be impaired in persons with this syndrome.^{114,115} Similarly, peak heart rate has been found to be approximately 85% that of the age-predicted maximum.¹¹⁰ Chronotropic incompetence and sympathetic dysfunction are often cited as being the main contributory factors to these effects.^{116–119} Abnormalities of the atrioventricular node may also be associated with dysrhythmias and heart block of variable degrees. Pulmonary capacity and air flow dynamics may also be affected, because pulmonary dysplasia and relative upper airway obstruction have been described in this population.¹¹⁰ In contrast to persons with Down syndrome, persons with fragile X syndrome may possess relatively increased levels of autonomic tone with possible consequences on heart rate during exercise and the recovery period. However, the effect that this characteristic has on exercise and performance is unclear.^{120–122}

Medical Considerations

Musculoskeletal Injuries

Common musculoskeletal conditions in persons with Down syndrome or fragile X syndrome include strains, sprains, contusions, poor muscle tone, generalized joint laxity, and a variety of malalignments, including scoliosis, pes planus/cavus, hallux varus/valgus, and hip instability.^{110,123–127} Although good data on injury rates and types of injuries are lacking, these conditions likely increase the risk of injuries such as patellofemoral pain syndrome, patellofemoral dislocation, spine pain, slipped capital femoral epiphysis, tendinopathies, and osteoarthritis.^{110,123,124} Knee injuries are most common, with track and field accounting for

the most sport-related injuries in total at state, national, and international events.¹¹⁰

Injury rates for 1000 participant-hours have been reported at 0.4 for Special Olympics and 2.0 for special education high school students.¹¹⁰ Special Olympics athletes in Texas with Down syndrome had a relative risk of injury or illness 3.2 times greater than other athletes.¹²⁸

An issue that demands attention is the presence of atlantoaxial instability in athletes with Down syndrome. These athletes are at risk of spontaneous or traumatic subluxation of the cervical spine and are required by most organizations to undergo a physical examination and flexion/extension radiographs of the cervical spine. Although a large majority of persons with Down syndrome who have radiographic instability (>4 to 5 mm of odontoid-atlas separation) are asymptomatic, all such athletes should be restricted from participation in sports activities that pose a risk for neck injury or inherently require excessive neck flexion/extension. Sports and events with high risk include, but are not limited to, pentathlon, diving, butterfly swim (and no flip turns no matter the stroke), high jump, gymnastics, soccer, judo, snowboarding, and alpine skiing.¹¹⁰ Atlantoaxial instability (AAI) can progress until skeletal maturity, and repeat evaluations throughout skeletal maturity may be necessary.¹¹⁰

Visual Impairment

Ocular and visual impairments are common in athletes with intellectual disabilities, with up to 50% afflicted by blepharitis, astigmatism, nystagmus, strabismus, lens opacities, and refractive errors.¹²⁹ Incorrect or delayed diagnosis and management of visual impairments may decrease coordination and motor skills, impair performance, and increase risk for injury.¹³⁰ The use of approved protective eyewear is strongly recommended, particularly during participation in high-risk sports and in those athletes with one functional eye.^{131,132} Further details are discussed in the Visually Impaired Athletes section of this chapter.

Epilepsy/Seizures

Epilepsy has been found to occur in persons with several conditions that are common in this category, including Down syndrome (up to 13%), autism disorders (20%), and fragile X syndrome.^{133–135} Further details are discussed in the Athletes with Cerebral Palsy section of this chapter.

Cardiac Abnormalities

Approximately half of all persons with Down syndrome have congenital heart problems, most often an endocardial cushion defect, followed by isolated septal defects.¹¹⁰ Stenosis in surgically corrected valves or subaortic stenosis may complicate prior repair of an atrioventricular canal defect, and mitral and aortic regurgitation may present primarily without prior congenital heart problems. As a result of flawed connective tissue, persons with fragile X syndrome may have mitral valve prolapse and aortic root dilatation, which particularly occur in mature individuals. Although not well described in this population, these heart conditions have exercise implications similar to those in persons with Marfan or Ehlers-Danlos syndromes. Clinicians should obtain a detailed history to screen for significant cardiovascular

problems, including syncope, lightheadedness, and dyspnea. Furthermore, a thorough clinical examination to evaluate for the presence of aortic regurgitation and mitral valve prolapse is critical. Lastly, detailed cardiac testing including stress testing and echocardiography may be necessary.

Adaptive Equipment

Intellectually disabled athletes may benefit from simple variations of standard equipment; for example, balance beams may be wider than standard size to accommodate impaired visual acuity and balance, balls may be larger and softer, and equipment may have straps and larger handles to help with grip. The athletes may benefit from assistance from other persons as well, such as with the use of tethers for alpine skiing.

VISUALLY IMPAIRED ATHLETES

Visually impaired athletes may possess a number of different conditions that limit visual acuity. In the Paralympics, these athletes are separated into categories based on visual acuity for some events, or they may compete directly against each other regardless of their amount of remaining sight. In the latter case, athletes are often required to wear “black-out” masks to reduce any possible competitive advantage based on sight.⁹

Exercise Physiology

Cardiopulmonary physiology of visually impaired athletes is similar to that of able-bodied athletes.¹³⁶ However, other associated syndromes, as previously described in various groups, may increase risk of cardiopulmonary abnormalities.

Medical Considerations

Musculoskeletal Injuries

Athletes with visual impairments may be particularly prone to acute injuries of the distal lower extremities such as ankle sprains and bruises. These injuries result from collisions and falls during competition in unfamiliar environments wherein adequate guidance is lacking.¹⁰² Visually impaired athletes also may be at risk for chronic overuse injuries due to altered biomechanics, induced by poor proprioception, and loss of the visual component of balance.⁷⁰

Flight Dysrhythmia

The visual input of light is an exogenous time cue for the body and for adjustments to the circadian rhythm that governs the sleep-wake cycle, core temperature, and other key aspects of bodily adjustment. Rapid desynchronization of the circadian rhythm, as noted with transmeridian travel, may result in flight dysrhythmia, commonly known as “jet lag.” This may affect sleep characteristics, cognition, athletic performance, and risk of heat-related illness.¹³⁷ Loss of light as an exogenous cue in persons with visual impairment may result in a diminished physiologic response to flight dysrhythmia.¹³⁸ This diminished response may have a negative impact on athletes who are visually impaired as they plan for international travel in an effort to allow for time for synchronization to improve performance and reduce risk for injury.

Adaptive Equipment

Although some sports such as judo require minimal interventions, some events may make use of companions as guides. For example, during track events, some companions may provide auditory cues or tap when athletes are in danger of a lane violation. In other events such as goalball, companions may use a sound-based modification in which the ball has internal bells, requiring all participants, coaches, and spectators to remain quiet to allow for auditory tracking of the ball. Other modifications may include the use of bright light, course and equipment colors with high levels of contrast, heavily padded ropes and boundaries, and guide rails.

LES AUTRES

Athletes in the Les Autres group do not fit into one of the other five categories. This category encompasses a wide range of different medical conditions of at least some familiarity to most clinicians, including multiple sclerosis, osteogenesis imperfecta, muscular dystrophy, and dwarfism. However, this group potentially includes an enormous range of conditions and syndromes, with 1 of the 10 types of permanent impairments, listed in the “Classification” section of this chapter.

It is imperative that the clinician be familiar with other features of a given syndrome that may affect sports performance, safety, and overall health. Although an in-depth discussion of the various conditions is beyond the scope of this chapter, the clinician may draw upon some of the principles presented in the prior sections of this chapter. For example, aspects that warrant assessment include but are not limited to cardiopulmonary, thermoregulatory, cognitive, restricted ROM or motor functions, seizure risk, and adaptive equipment. A clinician may then draw upon this assessment and other knowledge of the person's condition to make informed recommendations regarding the choice of sport, safety, and precompetition planning.

SUMMARY

Medical conditions that affect athletes with special needs are numerous and diverse in clinical presentation, enhance risk for injury and illness, and can affect peak athletic performance. Yet given the plethora of benefits from regular physical activity, increased participation in competition, and the exploration and expansion of boundaries of sport in this population, it is important for clinicians to be familiar with the different “athlete classes” and their clinical characteristics. It is only then that clinicians will be able to perform comprehensive preparticipation physical examinations, provide quality care, match patients to appropriate sports, counsel on risk reduction for injury and illness, and appropriately refer athletes to consultants and allied health professionals to provide the best medical care.

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Citation:

Slocum C, Blauwet C, Allen JA. Sports medicine considerations for the Paralympic athlete. *Curr Phys Med Rehabil Rep*. 2015;3(1):25–35.

Level of Evidence:

V

Summary:

A brief overview of the common athlete classifications and injuries seen in the para-athlete.

Citation:

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Level of Evidence:

V

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Level of Evidence:

V

Summary:

A brief evidence-based review of the issues contained in this chapter, as well as psychosocial and assessment considerations as they specifically pertain to the pediatric disabled athlete population.

Citation:

Burkett B. Technology in Paralympic sport: performance enhancement or essential for performance? *Br J Sports Med*. 2010;44(3):215–220.

Level of Evidence:

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Citation:

Lynch JH. The athlete with intellectual disabilities. In: O'Connor FG, ed. *ACSM's Sports Medicine: A Comprehensive Review*. Wolters Kluwer, Lippincott Williams and Wilkins; 2013.

Level of Evidence:

V

Summary:

A chapter within a highly used sports medicine textbook, with a brief summary of key points for the medical provider when caring for the intellectually disabled athlete.

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